LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 11 Test (Set B): Electricity

Chapter / Topic: Electricity

For teacher use only. Do not distribute to students.

SECTION A

Q1. The SI unit of potential difference is the: **[1]**

Answer: (d) volt

Q2. The length of a wire is doubled and its area of cross-section is halved. Its resistivity: **[1]**

Answer: (b) stays the same (resistivity depends only on the material and temperature)

Q3. Two 4 Ω resistors are connected first in series and then in parallel. The ratio of the series resistance to the parallel resistance is: **[1]**

Answer: (c) 4. Series 8 Ω, parallel 2 Ω.

Q4. How much electrical energy does a 100 W bulb use in 10 hours? **[1]**

Answer: (b) 1 kWh. E = 0.1 kW × 10 h

Q5. The filament of an electric bulb is made of: **[1]**

Answer: (c) tungsten

Q6. **Assertion (A):** Alloys are used in electric heating devices. **Reason (R):** Alloys have a higher resistivity than pure metals and do not oxidise easily… **[1]**

Answer: (a)

SECTION B

Q7. A current of 0.5 A flows through a bulb for 10 minutes. How much charge flows through it? **[2]**

Answer: Q = It = 0.5 A × (10 × 60) s = 300 C

- Formula and time in seconds: 1
- 300 C: 1

OR, answer: n = Q/e = $1/(1.6 \times 10^{-19}) = 6.25 \times 10^{18}$ electrons

- Formula: 1
- 6.25×10^{18}: 1

Q8. Why does the cord of an electric heater not glow, while its heating element does? **[2]**

Answer: The heating element is made of an alloy with a high resistance, so a lot of heat is produced in it ($H = I^2Rt$) and it glows. The cord is made of copper, which has a very low resistance, so very little heat is produced in it.

- Element has high resistance: 1
- Cord has low resistance: 1

SECTION C

Q9. Resistors of 2 Ω, 3 Ω and 5 Ω are connected in series to a 10 V battery. **[3]**

(i) R = 2 + 3 + 5 = 10 Ω [1]
(ii) I = V/R = 10/10 = 1 A [1]
(iii) V = IR = 1 × 5 = 5 V [1]

Q10. A current of 2 A flows through a 50 Ω resistor for 1 minute. Calculate the heat produced. How would the heat change if the current were doubled? **[3]**

Answer: $H = I^2Rt = 2^2 \times 50 \times 60 = 12000$ J (1.2×10^4 J). If the current is doubled, the heat becomes four times (48000 J), because $H \propto I^2$.

- Formula: 1
- 12000 J: 1
- Four times: 1

OR, answer: I = P/V = 1500/220 ≈ 6.8 A. R = $V^2/P = 220^2/1500$ ≈ 32.3 Ω. No: it draws more than 5 A, so the fuse would blow. It needs the 15 A circuit.

- Current: 1
- Resistance: 1
- Not safe, with reason: 1

SECTION D

Q11. Answer the following: **[5]**

(i) At constant temperature, the current through a conductor is directly proportional to the potential difference across its ends (V = IR). Experiment: connect a resistor (nichrome wire), an ammeter in series, a battery, a key and a rheostat, with a voltmeter in parallel across the resistor. Change the current using the rheostat (or the number of cells) and note V and I each time. V/I stays constant, and the V–I graph is a straight line through the origin. [3]

- Statement: 1
- Circuit: 1
- Observation (V/I constant, straight line): 1

(ii) The slope (V/I) gives the resistance of the conductor. The steeper line represents the larger resistance. [2]

- Slope = R: 1
- Steeper = larger R: 1

OR, answer: (a) (i) I = 6/(1 + 2) = 2 A, so $P = I^2R = 2^2 \times 2 = 8$ W. (ii) The voltage across the 2 Ω resistor is 4 V, so $P = V^2/R = 16/2 = 8$ W. The power is the same (8 W) in both cases. (b) In series, the current is the same through every appliance, and the voltage is shared, so appliances don't get their full rated voltage. If one appliance fails, the whole circuit breaks, and appliances cannot be switched on or off independently.

- (a) (i) 1, (ii) 1, comparison 1
- (b) two reasons: 1 + 1

SECTION E

Q12. A family uses five 10 W LED bulbs for 5 hours a day, two 75 W fans for 8 hours a day, and a refrigerator rated 150 W that effectively runs for 10… **[4]**

(i) 5 × 10 W × 5 h = 250 Wh = 0.25 kWh [1]

(ii) Bulbs 0.25 kWh + fans 2 × 75 W × 8 h = 1.2 kWh + refrigerator 150 W × 10 h = 1.5 kWh, total 2.95 kWh [1]

(iii) Energy = 2.95 × 30 = 88.5 kWh. Bill = 88.5 × ₹5 = ₹442.50. Ways to reduce it (any one): switch off fans and lights when not needed; use energy-efficient (star-rated) appliances; keep the refrigerator door closed. [2]

- ₹442.50: 1
- One sensible way: 1